

Quantitative Metrics for Decomposing Timing and Magnitude Differences in Influenza Forecasts

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Introduction

Seasonal respiratory viruses like influenza, COVID-19, and RSV place a significant burden on public health systems, making accurate epidemic forecasting crucial. Specifically, forecasting at the local level could offer greater geographic precision than the state-level but only where there is a meaningful divergence, often found in certain geographic areas.

To quantify this divergence, conventional error metrics such as Root Mean Squared Error (RMSE) are often used but cannot distinguish whether differences arise from timing or magnitude, which have fundamentally different public health implications.

On the other hand, curve-based similarity metrics offer a more interpretable alternative. Discrete Fréchet distance captures the worst-case pointwise separation between two curves, making it sensitive to sharp mismatches, while Dynamic Time Warping (DTW) uses non-linear temporal alignment to decouple timing shifts from magnitude differences. Together, Fréchet distance and DTW-derived measures like signed lag and magnitude ratio preserve directional and structural information about how and where forecasts diverge, enabling more actionable insight for public health decision-making.

Research Goal

This project aims to identify the best metric for decomposing timing and magnitude divergence between local- and state-level respiratory virus forecasts. Using influenza, we evaluate candidate metrics to see if they capture specific forecast features. Findings are expected to generalize to COVID-19 and RSV.

Methods

Discrete Fréchet Distance

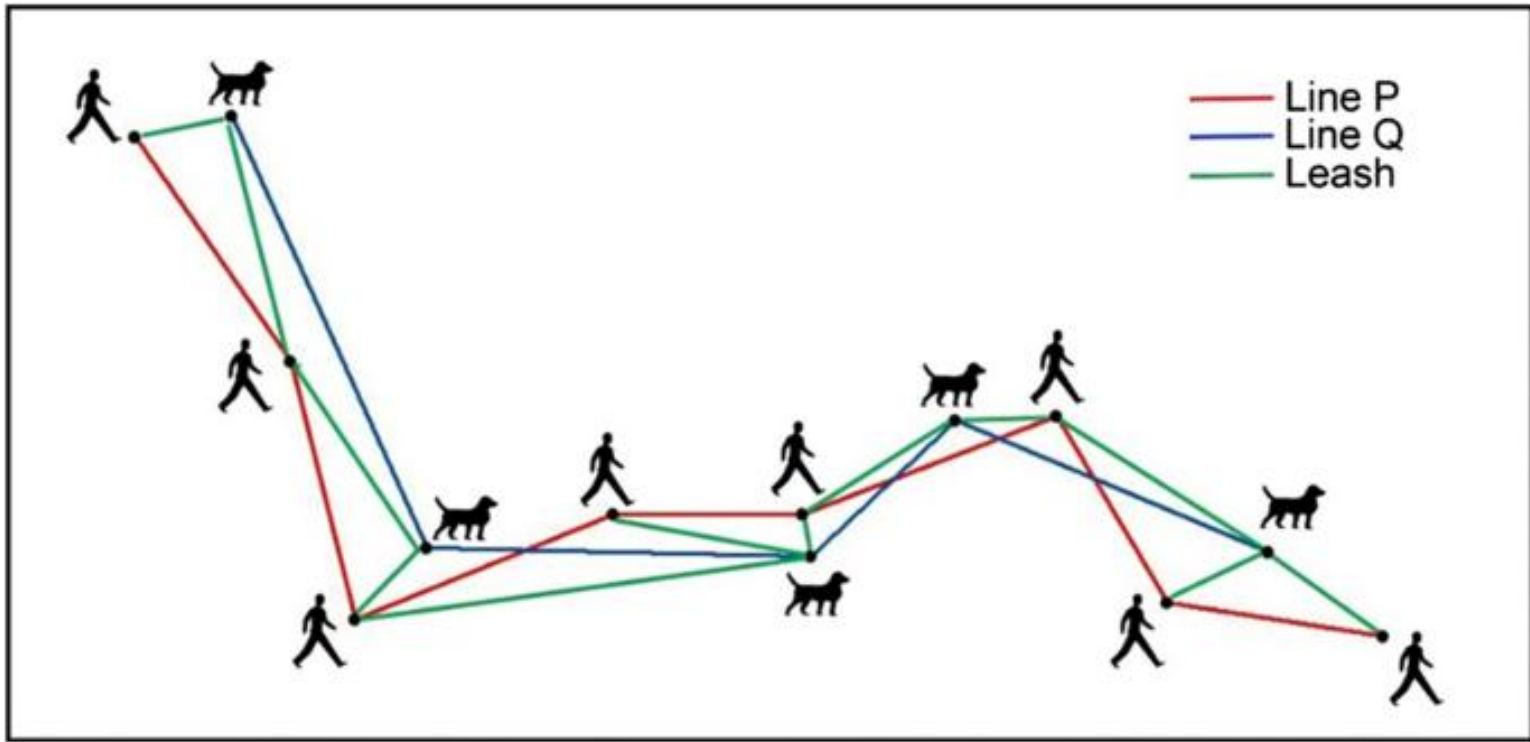
This metric can be visualized by thinking about a dog and its owner walking on separate curves. The Fréchet distance is the shortest leash that lets both complete their paths, logged at each time step.

Dynamic Time Warping

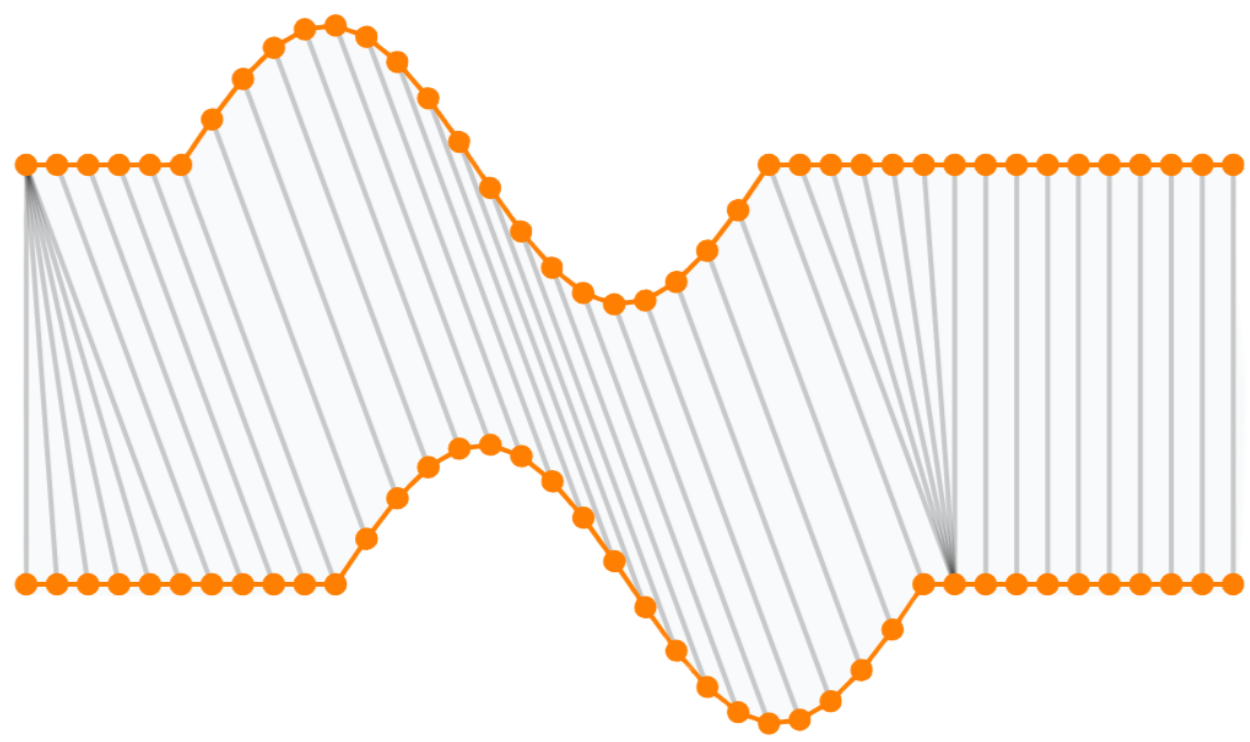
This metric can be visualized as two people walking the same trail at different speeds. DTW finds the optimal way to sync up their steps to minimize total paired distance.

$$C = \langle (i_1, j_1), \dots, (i_k, j_k) \rangle$$
$$d_F(P, Q) = \min_C \max_{(i,j) \in C} d(p_i, q_j)$$

$$\gamma(i, j) = d(p_i, q_j) + \min \begin{cases} \gamma(i-1, j) \\ \gamma(i, j-1) \\ \gamma(i-1, j-1) \end{cases}$$
$$\text{DTW}(P, Q) = \gamma(m, n)$$



Source: doi.org/10.1007/s12518-024-00605-3



Source: Romain Tavenard

Figures and Results

To validate that DTW signed lag and Fréchet distance capture what they claim to measure, we examine two contrasting Texas Health Service Areas: Amarillo, a more isolated and rural county, and Dallas, an urban county, to investigate the effect of urbanization on local vs. state differences. Furthermore, the below scatter plots confirm that signed lag correlates strongly with true centroid timing lag and Fréchet distance correlates strongly with true magnitude ratio across all three windows.

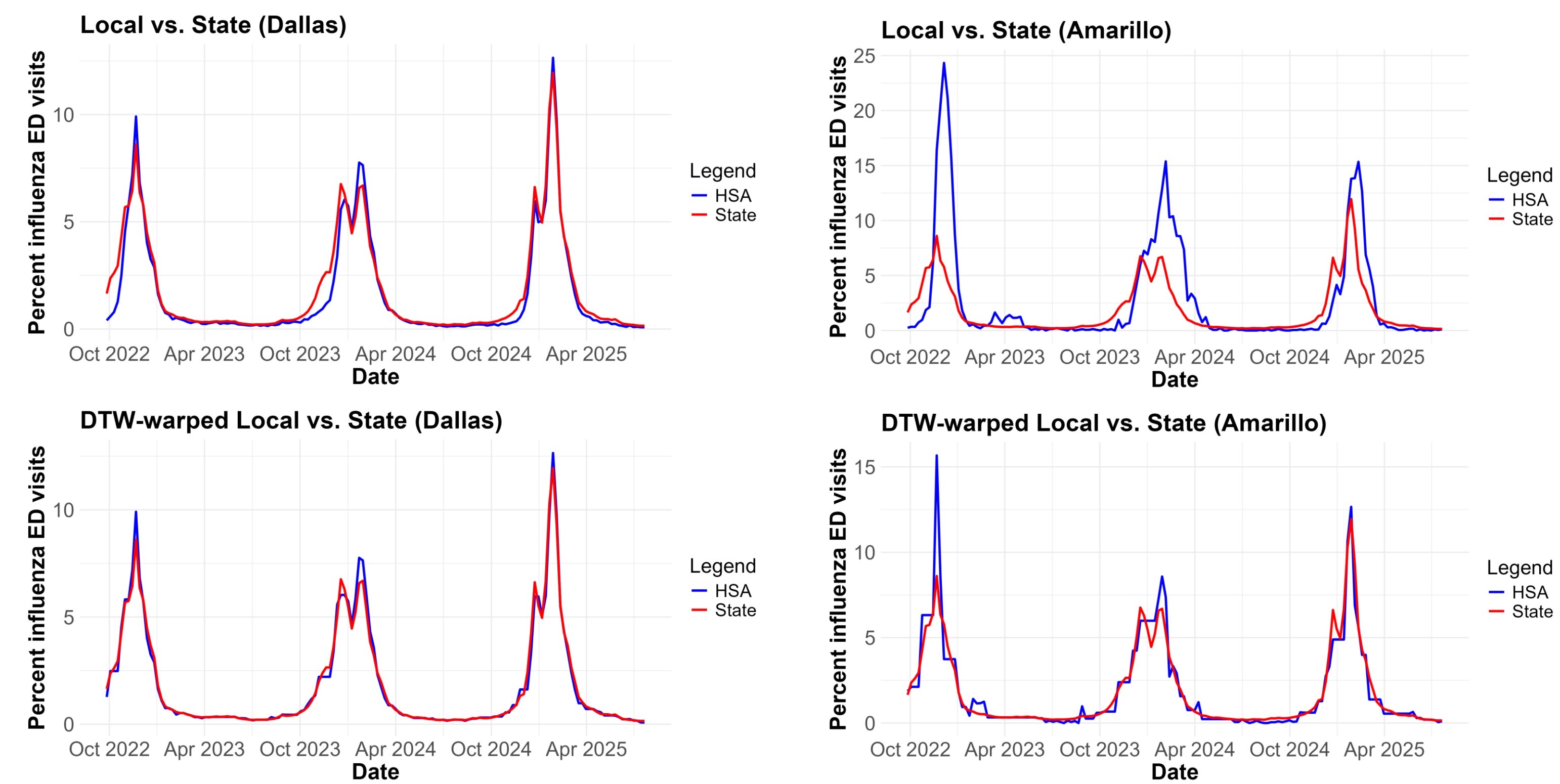


Figure 1. Local vs. state-level influenza emergency department visit curves for Amarillo and Dallas HSAs across three epidemic windows.

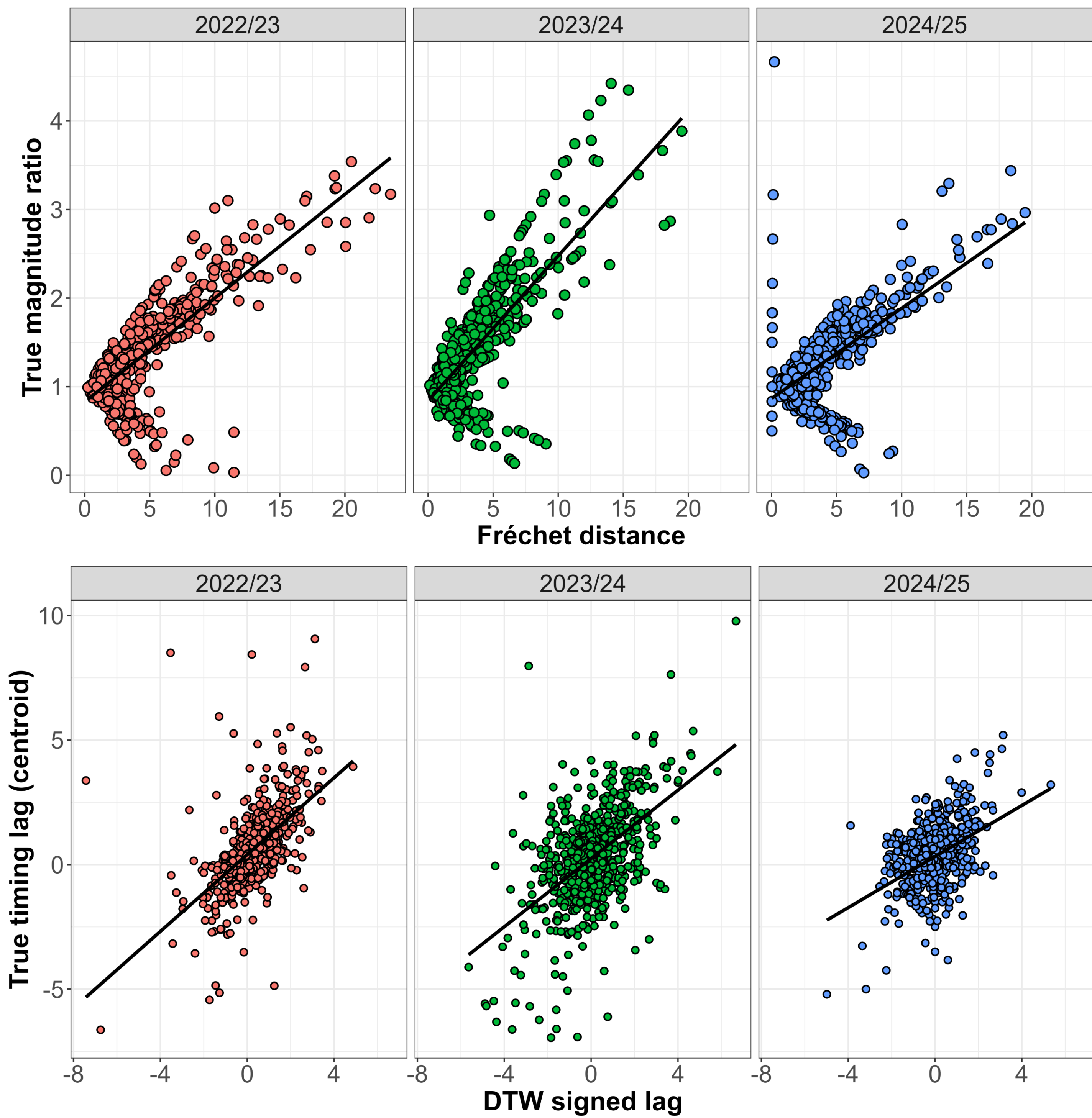


Figure 2. DTW signed lag vs. true centroid timing lag (left) and Fréchet distance vs. true magnitude ratio (right), faceted by epidemic window.

Conclusion

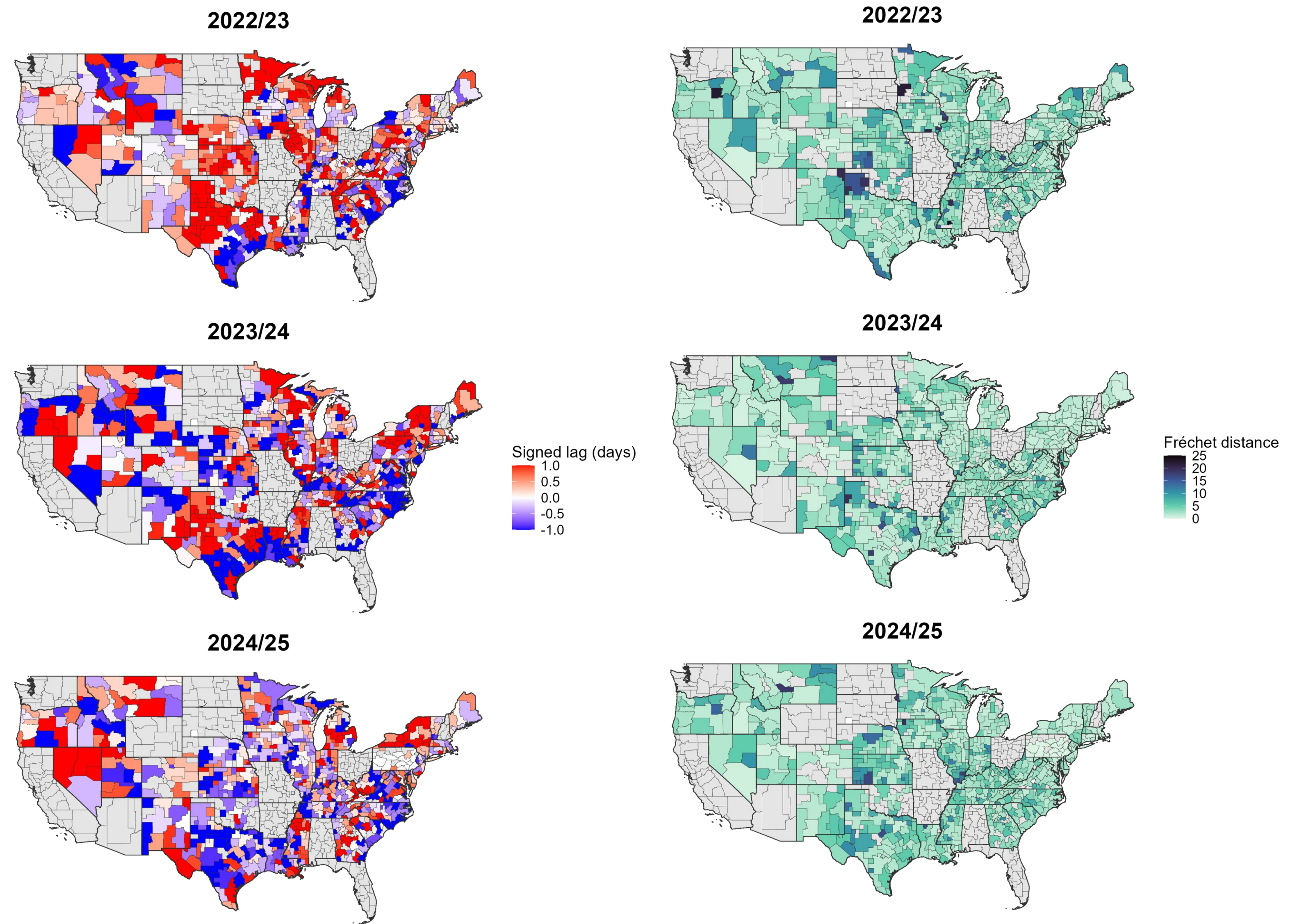


Figure 3. Choropleth graphs of DTW signed lag (left) and Fréchet distance (right) for HSAs

In the US choropleth graphs above, DTW signed lag captures directional timing divergence between local and state influenza forecasts, outperforming RMSE by preserving the sign and magnitude of the shift. Signed lag also has a bit of a geographic component which changes window to window. Discrete Fréchet distance complements signed lag by contextualizing the magnitude differences, flagging HSAs where local and state curves diverge sharply. For Fréchet distances, most regions have similar values, with some outliers concentrated in certain regions. These metrics lay the groundwork for composite forecast measures that capture timing and magnitude to check forecast quality and similarity to true values.

Acknowledgments

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